

Smart Recycle Bin

DESIGN THINKING FOR AI SOLUTIONS

Group 7

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1.Introduction

In the era of rapid urbanization and increasing environmental awareness, traditional waste management systems are no longer sufficient. Fixed collection routes, manual sorting processes, and a lack of real time data contribute to inefficiencies that result in fuel overuse, overflowing bins, low recycling rates, and rising operational costs. These outdated practices also hinder compliance with growing environmental regulations, especially in industries such as food service, where strict waste sorting is enforced. The demand for intelligent, sustainable, and data-driven solutions in urban waste management has never been more critical.

This project addresses that need through the development of an AI-powered Smart Waste Management System, designed to enhance both waste sorting and collection processes. By integrating image classification and sensor technologies, the system can accurately identify waste types and monitor bin fill levels in real time. IT provides immediate sorting guidance to users, generates automated performance reports for managers, and sends alerts when bins are full or misused. The solution not only supports fast food restaurants in maintaining hygiene and compliance but also has the potential to scale for municipal use, optimizing collection routes and reducing environmental impact.

By operating within the subspace of AI-powered smart waste sorting and collection optimization, this project aligns with the broader goal of building efficient, eco-friendly urban infrastructures. The result is a smarter, cleaner and more sustainable approach to waste management-benefiting both business operations and the environment.



2.Data Source

Public & have

Our project, which focuses on the implementation of smart bins for efficient urban waste management, relies on publicly available datasets that are already in our possession. These data sources are essential for understanding waste patterns and supporting sustainable development.

- **Toronto Waste Management Data (Solid Waste Tonnages):**
Provides solid waste tonnage data crucial for understanding waste trends in urban areas. This dataset helps us analyze how much waste is generated in different parts of the city, allowing our smart bins to be placed more strategically. It also assists in identifying peak waste periods, which helps optimize collection schedules and reduce overflow issues. This information is vital for the smart bin system to operate efficiently and respond to real-time waste disposal needs.
- **OpenAI Environment & Sustainability Datasets:**
Offers insights into broader environmental impacts and sustainable practices. These datasets help shape the overall environmental framework of our project. By incorporating this data, our smart bins are designed not only to collect waste but to promote sustainability through features like waste type detection, data monitoring, and smart alerts. The datasets also guide us in integrating practices that align with global sustainability goals, contributing to greener urban environments.

These sources play a foundational role in the planning and execution of our smart bin initiative. They ensure that our project is informed by real data, supporting practical deployment and long-term environmental benefits.

Public & Need

- **Real-time waste generation statistics from Toronto Open Data (if available):**
This data is essential to monitor current waste generation patterns in urban areas. It supports the smart bin system by enabling responsive waste collection and trend analysis for better planning.
- **Waste generation trends from GODAN (Global Open Data for Agriculture and Nutrition):**
Offers valuable insights into global waste production behaviors. This helps align our smart bin project with sustainable practices and global environmental goals.

Private & Need:

- **User interaction data with smart bin (how/when it's used):**
Necessary for understanding human behavior around waste disposal. This data helps improve smart bin features, such as usage prediction, automated alerts, and user engagement tracking.
- **Data on types of non-recyclable items detected:**
Crucial for training the smart bin's recognition system. This helps in sorting waste more accurately and enhances the bin's ability to promote recycling by identifying common non-recyclables.
- **Waste management in the U.S | Statista:**
Provides comparative insights into waste management practices and volumes in

other regions. This aids in benchmarking the effectiveness of our smart bin system and identifying best practices from other countries.

User & Nice to have:

- **User feedback data:**
Valuable for enhancing user experience and improving the functionality of smart bins based on actual user suggestions, complaints, and behavioral patterns.

Public & Nice to have:

- **Global Environmental Impact Data (Pollution & Sustainability Metrics from OpenAI Datasets):**
Offers context on broader environmental concerns. These insights help align the smart bin project with global sustainability efforts and inform eco-conscious design choices.
- **Smart city waste management reports:**
Provide examples and benchmarks from other smart city initiatives. This helps evaluate and refine our smart bin system to meet urban efficiency and sustainability standards.

Private & Nice to have:

- **Energy consumption of smart bin:**
Useful for assessing operational efficiency and identifying opportunities to reduce energy usage in our smart bins, making them more sustainable.
- **Real-time weight measurement of waste per bin:**
Helps track waste disposal patterns more precisely and enhances automated collection systems by detecting fill levels and scheduling pickups accordingly.

3. Target Audience and Use Case Scenario

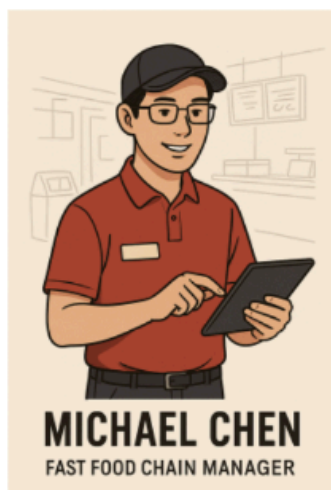
Our target audience includes both stakeholders, such as commercial establishment managers and waste management decision makers, and regular customers. The former are the individuals or entities to whom we will present the project, as they must be the first to recognize the value and usefulness of the new system we are offering. The latter represent the end users who will interact with the smart bins on a daily basis. This project has been designed around the needs of both groups.

To ensure optimal use of the smart bin system, the ideal scenario would be locations that generate a high volume and variety of solid waste. Environments such as shopping malls and airports fit this profile particularly well due to their large foot traffic and diverse waste generated. In these spaces, the dining or food court area are our starting focus points. It is not only where the majority of waste is produced, but also where the proper classification of materials, such as organic, recyclable, and general waste, becomes most crucial.

Proper waste separation determines the value of the recyclables, which are typically sold by weight. However, even a single misplaced item, like an unfinished soda bottle or cup disposed of in the wrong bin, can contaminate an entire batch. Once contaminated, these materials may lose their potential for reuse and may no longer be usable for recycling purposes.

With this in mind, fast food chains will be our initial implementation focus. These establishments not only produce consistent volumes of solid waste across multiple locations, but they also offer standardized layouts that allow for repeatable, scalable testing. By starting with a fast food chain, we aim to generate measurable impact quickly while refining the solution for broader applications, such as the scenarios previously mentioned.

Persona: Michael Chen – Fast Food Chain Manager



- Checks restaurant cleanliness and hygiene daily
- Monitors waste management processes to ensure compliance
- Regularly submits operational and environmental reports to headquarters
- Interested in new technologies but focuses on practical results and return on investment (ROI)

To make our target audience more tangible, let us introduce **Michael Chen**, the manager of a fast-food chain restaurant. Michael represents a typical stakeholder in the implementation of our system. He is responsible for maintaining restaurant cleanliness and hygiene, and he closely monitors waste management procedures to ensure regulatory compliance.

Part of his routine includes submitting both operational and environmental reports. As someone who is familiar with the importance of standardization, Michael would like new ways to automate processes and reduce inefficiencies, especially in areas like waste sorting that are not given enough importance.

Michael's main goal is to **minimize waste sorting errors** that could result in customer complaints or even fines due to improper disposal. He's also interested in having access to real-time data and automated reporting tools that help to optimize operations and improve communication with the waste management partner companies.

At the same time, he is also interested in implementing new technologies, aiming at getting practical results. His focus is on **return on investment (ROI)** and **long term feasibility**. He worries about the **upfront costs** of adopting the new system and the potential challenges it could introduce into the operations.

Our smart bin solution is built with people like Michael in mind. It offers:

- A user-friendly interface that staff can quickly learn to use.
- Real time classification data and alert systems that help reduce human error.
- Integration with reporting tools to enhance the communication with waste collectors and gain insights of the higher residues generated in the establishment.
- A setup that reduces installation costs and allows for scalable expansion, such as facing scenarios where new kinds of products (waste) are implemented in the establishment.

Michael is not just a fictional persona, he represents the needs of real decision makers in fast paced environments. Designing with his expectations in mind ensures that our solution aligns with business priorities while supporting environmental impact goals.

4. AI Intents

The system will be interacting frequently with many different users while categorizing wastes. Besides focusing on AI classifications, the user experience must be easy, fast and pleasant to the user.

Recommend with confidence

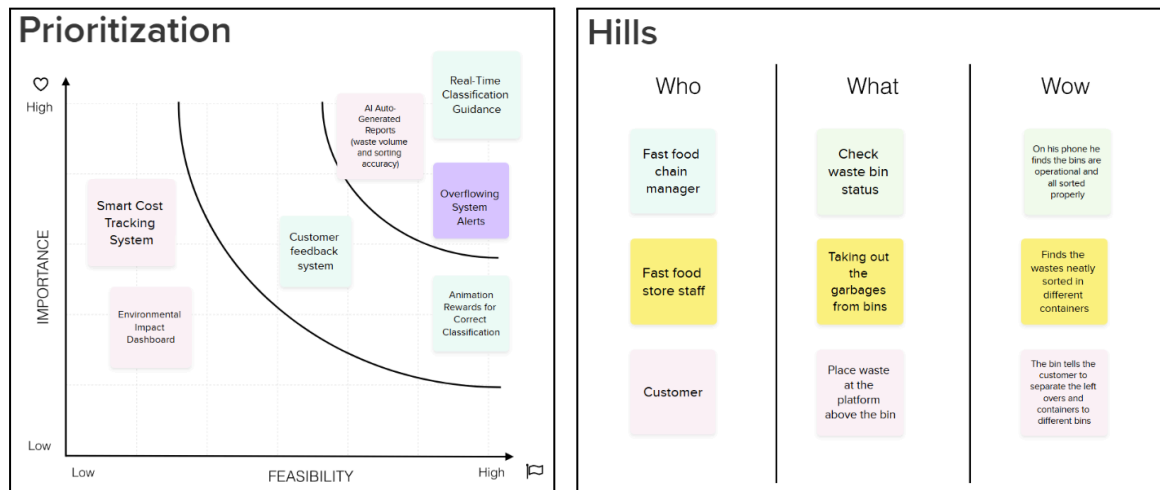
- Correct waste classification.
- Classifications according to the bins which should be divided in at list 3 principal spots: recyclable, organic, waste.
- Periodic retraining with new data.
- Direct to waste bin when the classification is of low confidence.
- False classifications can be corrected via user feedback on screen for further training.
- A primary model for each industry is trained while strategically utilizing models from other industries.

Enrich your interactions

- Direct people where to drop the trash according to classification result.
- Non-disruptive pleasant sound when placed in correct bin
- Warnings to management of problems such as malfunctioning and overflowing.
- Rewards enabled by mobile app to encourage consumers practicing correct disposal.

5. Smart Bin Feature Prioritization

Based on our analysis of user insights derived from Hills and the Big Idea Vignettes, we have prioritized smart bin features focusing on their importance and feasibility. This ensures our Minimum Viable Product (MVP) meets immediate user needs effectively.



Prioritized Features

Real-Time Classification Guidance (High Importance, High Feasibility)

Description: Real-time camera recognition provides users with instant visual guidance for correctly disposing of different types of waste.

Rationale: Directly addresses our Hill, guiding customers effectively at the moment of waste disposal to ensure accurate sorting.

Overflowing System Alerts (High Importance, High Feasibility)

Description: Automated alerts when the bin reaches 80% capacity, notifying management to prevent overflow.

Rationale: Aligns perfectly with our second Hill by enabling managers to monitor bin status remotely, ensuring operational continuity.

AI Auto-Generated Reports (Moderate to High Importance, Moderate to High Feasibility)

Description: Automatic weekly/monthly reports on waste volumes and sorting accuracy are sent directly to management.

Rationale: Facilitates managerial oversight and operational optimization, further supporting the monitoring of Hill and ensuring consistent improvements.

Additional Considerations for Future Development:

- **Animation Rewards for Correct Classification:** Although feasible, this feature holds lower immediate priority compared to guidance and operational stability.
- **Smart Cost Tracking System and Environmental Impact Dashboard:** These features, while valuable, represent a lower immediate impact relative to our prioritized items. However, they provide excellent opportunities for future enhancements.

Our prioritization clearly identifies Real-Time Classification Guidance and Overflowing System Alerts as critical MVP features, aligning closely with user needs defined by our Hills and Big Idea insights. Implementing these will significantly improve waste sorting accuracy and operational efficiency.

6.To-Be Scenario Map

The TO-BE scenario shows a future-oriented workflow in which Michael, a fast-food restaurant manager, and his team use AI-powered smart bins to improve waste management, increase sustainability, and assure compliance with food safety regulations. The goal is to show how this intelligent system transforms routine tasks into data-driven, environmentally conscious processes.

Step 1: Remote monitoring of bins

Michael begins his day by remotely accessing the "AI Waste Insights" dashboard on a tablet. This technology gives real-time information about bin fill levels and trash type. This allows the personnel to handle overflow more effectively and reduce wasteful collections. This enables managers to make timely, informed choices, resulting in more efficient operations and less manual supervision.

Step 2: AI-Assisted Sorting for Staff

Both new and existing staff benefit from an integrated trash sorting assistance. The AI tool automatically determines item kinds (organic, recyclable, or landfill) and directs the user accordingly. This process reduces errors, streamlines onboarding, and assures constant adherence to food safety and environmental norms.

Step 3: Tracking Sustainability Progress

The system delivers monthly sustainability reports that show waste reduction, recycling rates, and cost savings. These insights are automatically sent to the headquarters to demonstrate the restaurant's progress. This gives Michael and his team a sense of accomplishment while also contributing to the company's overall sustainability goals.

User Journey and Emotional Impact

The user journey progresses from data empowerment to confidence through simplified sorting, culminating in pride in meeting environmental goals. The smart bin system makes waste disposal a simple and gratifying experience for both managers and employees.

Alignment with Project Goals

This TO-BE scenario directly supports the smart bin initiative's broader goals of lowering operational inefficiencies, improving employee performance, and reaching sustainability requirements. It demonstrates a practical, real-world use of AI for social and environmental benefit.

7. Prototype Key Tasks

The Smart Bin prototype identifies distinct responsibilities for both customers and managers in a fast-food restaurant environment:

Customer Tasks

- Place waste on the counter above the bins so the system can determine each item's appropriate bin (organic, recyclable, or landfill).

Manager Tasks

- Monitor the status of bins remotely through the system interface.
- Address any issues flagged by the system, such as a full bin or a sorting error notification.

8. Usability Testing Scripts (Customer)

To evaluate the functionality and intuitiveness of the prototype, scenario-based scripts were developed for both customers and managers.

Customer Scenario

The customer is instructed to imagine finishing their meal at a fast-food restaurant and approaching the Smart Bin with waste. The following questions are used to assess usability:

1. Are you going to segregate the waste?
2. Do you know how the wastes are segregated?
3. How long is it going to take just to dispose of the waste?

These questions aim to assess awareness, clarity of the interface, and time efficiency from the user's perspective.

9.Low-Fidelity Paper Prototype Overview (Customer)

A low-fidelity paper prototype was created to visualise how a customer interacts with the Smart Bin system.

Testing Task

Participants are asked:

“You want to start disposing of waste into the bin. How can you use the prototype to ensure it’s properly disposed of?”

This task helps validate whether the prototype effectively communicates the disposal process to users and ensures proper interaction with the system.

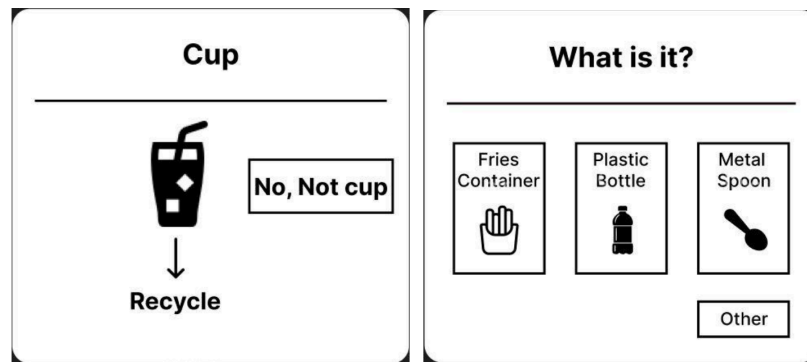
Customer Scenario Visual Flow

The prototype includes the following key elements to guide the user through the waste disposal process:

- A welcome interface labelled **"SMART BIN"** that displays available bin categories (e.g., Organic, Recyclable, Landfill).
- A visual instruction prompts the customer to **place the waste on the designated counter area**.



- A detection phase where the system **identifies the item** and displays a prediction of what it is.
- A final screen that **guides the user to the correct bin** based on the detection.
- If the prediction is incorrect, the user has the option to click a button labelled **"No, it's not [item]"**. This brings up a screen asking the customer to **select the correct item** from a list.



Note: Providing feedback is optional, but doing so will help improve the accuracy of the system's machine learning algorithm over time.



10. Usability Testing Scripts (Manager)

Manager Scenario

The manager is instructed to imagine overseeing operations in a fast-food restaurant equipped with the Smart Bin system. The following questions are used to assess usability:

1. Are you able to monitor how effectively customers are disposing of waste?
2. Do you understand how to access disposal data for performance tracking?
3. How confident are you in making decisions based on the bin's analytics?

These questions aim to assess usability from the manager's perspective, focusing on ease of system monitoring, data accessibility, and actionable insights.

11. Low-Fidelity Paper Prototype Overview (Manager)

Testing Task

Participants are asked:

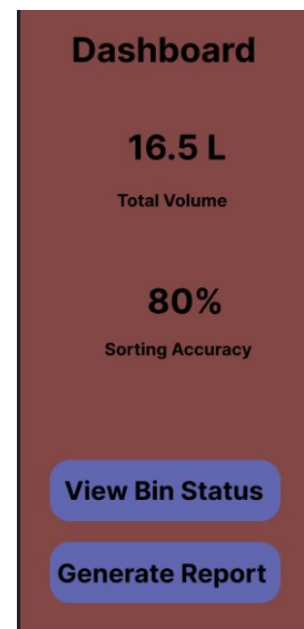
1. “You want to quickly check which bins in your store are close to overflowing or have classification issues. How would you do that using the dashboard?”
2. “You want to receive a notification when one of your bins has a sorting issue, and resolve the issue promptly at the right time and location. How can you use this prototype?”

These tasks help validate whether the prototype enables managers to quickly detect bin overflow or classification issues, receive timely alerts, and take appropriate action based on real-time system feedback and location-specific insights.

Manager Scenario Visual Flow

The prototype includes the following key elements to guide the manager through the monitoring and decision-making process:

- **Dashboard Overview:**
 - Displays key metrics such as **Total Volume (e.g., 16.5L)** and **Sorting Accuracy (e.g., 80%)**, giving a quick glance at overall performance.
 - Provides access to deeper insights via buttons: **“View Bin Status”** and **“Generate Report”**.



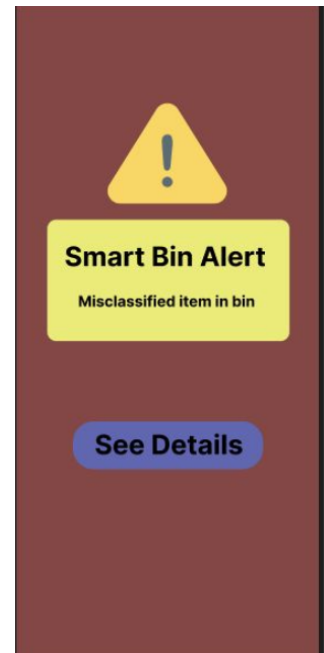
- **Bin Status View:**

- Shows individual bin fill levels across different categories (e.g., landfill, recyclable, compost), using clear percentage indicators (e.g., Compost Bin – 85%).
- Offers trend tracking through “**See Daily Trends**” and “**See Monthly Trends**” to help managers identify usage patterns.



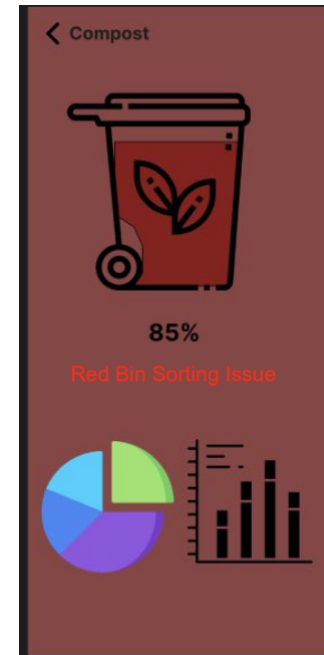
- **Smart Bin Alert System:**

- Automatically triggers a warning screen labeled “**Smart Bin Alert**” when a misclassified item is detected.
- A prominent “**See Details**” button directs the manager to further information about the issue.



- **Detailed Bin View with Issue Flagging:**

- Highlights specific bins with problems (e.g., **Red Bin Sorting Issue** at 85% capacity).
- Visual indicators (e.g., pie and bar charts) help analyze the type and frequency of misclassifications, enabling data-driven decisions.



This visual flow ensures that managers are alerted in real time, can diagnose specific bin issues swiftly, and have access to relevant usage and sorting data for operational adjustments and reporting. During usability testing, users reported that the interface was straightforward, intuitive and easy to use.

12. Conclusion

Reimagining Waste Management with AI

The Smart Recycle Bin project presents a transformative step toward modernizing waste management in urban environments. By leveraging AI-powered classification and sensor technologies, our system simplifies the act of waste disposal for customers and equips managers with real-time insights to oversee and optimize operations. Through intuitive interfaces, predictive alerts, and auto-generated reports, the Smart Bin enhances compliance, reduces contamination, and streamlines waste monitoring processes.

This solution is not only functional but scalable. It supports fast-food restaurants, where waste sorting accuracy and operational speed are critical, and can extend to larger municipal contexts. The Smart Bin promotes environmental sustainability by reducing human error and supporting effective recycling habits, while also offering measurable ROI through automation and data-driven decision-making.

Our design prioritizes user experience on both ends: ensuring customers can easily and correctly dispose of items, and enabling managers to resolve issues promptly, monitor bin status remotely, and track performance over time. From boosting staff confidence to

supporting broader sustainability goals, this system aligns with the evolving needs of smart cities and green infrastructures.

Next Steps

1. **Strengthen User Feedback Loops**
Implement simple ways for users and managers to correct misclassifications and provide feedback, helping the AI model improve over time.
2. **Pilot in Real-World Environments**
Deploy the system in selected fast-food locations to evaluate its performance, identify operational challenges, and refine the user experience.
3. **Incentivize Proper Sorting Behavior**
Introduce mobile-based rewards to encourage correct disposal habits and boost customer engagement through gamification.
4. **Expand Reporting & Analytics**
Develop more detailed dashboards and automated reports to support data-driven decision-making for managers and stakeholders.
5. **Build Strategic Partnerships**
Collaborate with local waste management providers to integrate our system into existing collection workflows and sustainability programs.